Unsupervised Learning Experiment Report

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***Abstract*— This report is a submission to the required assignment of the unsupervised Learning module, which contains the necessary contents about the experiments specified in the information document in Canvas.**

I. PROBLEM INTRODUCTION

A. Predict severity level with risk factors

The problem I choose for cancer dataset is to cluster and see the pattern of all data points based on patients' age, country, gender, other risk factors and stage of cancer, then I'll try to predict their severity level. I'll use unsupervised learning methods to find some patterns about the data, then assist it with some dimensionality reduction techniques. And in the end, I'll see if these techniques are help for in supervised learning.

B. Predict company bankruptcy based on financial data

This data set contains relatively complete data that illustrate the financial status of companies and their operational status, such as being bankrupt or not. I will use the same unsupervised learning and dimension reduction algorithms to explore the data, and then see how these methods affect the outcomes.

II. HYPOTHESES

A. Cancer Dataset

In cancer datasets, I first wished to identify the merits of the clustering algorithm. I believe the K-Means, or EM algorithm, can identify patient subpopulations hidden in cancer patient data, and that such subpopulations may be correlated with differing severities even if those subpopulations are imprecise or more nuanced in the initial diagnostic label. For the cluster effect, I anticipate the clustering effect of the EM algorithm on the worldwide cancer patient dataset will surpass that of K-Means due to the reality that the shape of the clustering may be irregular or overlapped.

Among the dimensionality reduction techniques, I believe that PCA does well at dimension reduction and preserving most of the variance in the original data, and is able to capture large amounts of clinical or biological variation in fewer dimensions. ICA can identify statistically independent "signals" or latent factors that can correspond to independent biological processes or disease drivers and can be more explanatory compared to PCA. In addition, I also hypothesize that datasets that have undergone PCA or ICA dimension reduction would provide neater and more meaningful cluster results and even improved generalization accuracy for neural network learners.

B. Bankruptcy Dataset

The clustering algorithm is executed on the bankruptcy dataset, and I anticipate that the K-Means or EM algorithm will be able to establish clusters of companies with different states of financial health (or risk of bankruptcy) from the company's data of bankruptcy. The clusters will reflect some degree of risk or operational stability in accordance with their financial characteristics. With the relative linearity of the financial values, I also venture that K-Means would be able to do relatively well in clustering on this data, especially in distinguishing groups with clear risk levels.

For the dimensionality reduction algorithm, I think the PCA would be capable of reducing the dataset's dimensions adequately, and the first principal components may be significant financial indicators that are able to explain much of the variance in a firm's financial health. I think that whereas random projection would show the most dramatic dimensionality reduction while processing this dataset, its integrity (especially in reconstruction errors) would not be as good as PCA. ICA would only work if there is an adequate number of independent components within the dataset. Bankruptcy data dimensionally reduced (specifically PCA or ICA) will improve the performance and efficacy of neural network classifiers in corporate bankruptcy prediction.

III. EXPERIMENTS ON CANCER DATASET

A. Data Exploration & Preprocessing

This data set contains global cancer patient data reported from 2015 to 2024, designed to simulate the key factors that influence cancer diagnosis, treatment, and survival. It includes a variety of features commonly studied in the medical field, such as age, gender, type of cancer, environmental factors, and lifestyle behaviors. This dataset contains 50000 data points and 15 columns.

Since I want to predict severity level, I have to divide the continuous 0 to 10 score into discrete boxes, so I evenly divided them into 3 levels, which are mild, moderate, and severe.

Besides, to prevent data leakage problem, I need to consider what features can be obtained before I know the label. We're not likely to know the cost of treatment before I really know how severe this cancer is, so it can not be in the features. And survival year is likely truncated because I don't know if the patients are alive or dead when they're reported, I don't think we're told about this background, so I tend not to use it. The rest of the variables, except for ID, can all be included as features.

I standardized all features. Each substracted their own means and divided by standard error.

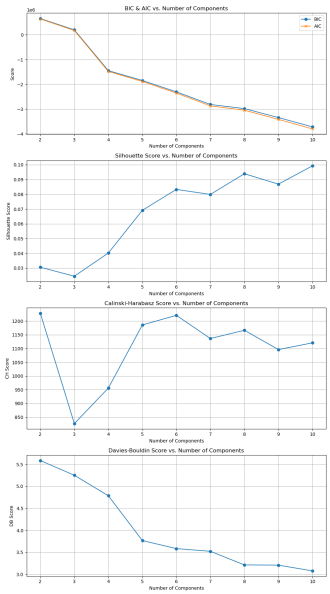

Fig. 1. 4 Criterions over Number of Components for EM

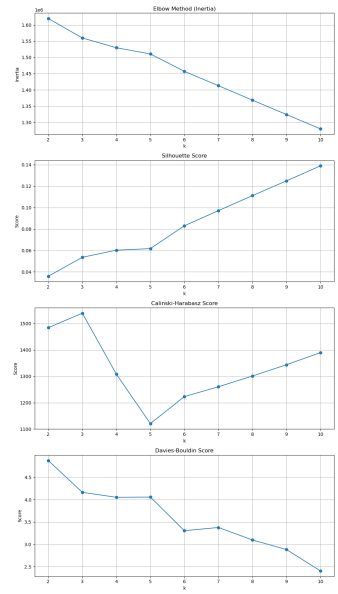

Fig. 2. 4 Criterions over k-value for K-Means

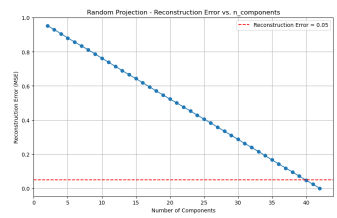

Fig. 3. Reconstruction Error vs. Number of Components for RP

B. Experiments on Clustering Algorithm

1) EM: I tried 4 criterion to test different number of components to recognize the pattern. The outcomes are shown in Fig.1.

(1) BIC: BIC is a criterion used for model selection, especially in Gaussian mixture models (GMM). It measures how well the model fits the data and penalizes the complexity of the model (i.e., the number of clusters). Normally I believe the smaller it is the better.

(2) AIC: Similar to BIC, is a model selection criterion that is also used to measure the trade-off between goodness-of-fit and complexity of a model. It doesn't penal as much of model complexity as BIC.

(3) The Silhouette Score measures how tight the samples within the cluster are and how separated the samples between the clusters are. It has a value between -1 and 1:

- Approaching 1 indicates that the sample is very similar to the objects in its own cluster and is far away from other clusters.

- Approaching 0 indicates that the sample is at the boundary of the two clusters.

- A negative value indicates that the sample may have been assigned to the wrong cluster.

(4) The Calinski-Harabasz Score: Also known as the variance ratio criterion, it evaluates clustering performance by calculating the ratio of intra-cluster dispersion to inter-cluster dispersion. Normally the larger it is the better. If it is largest at k=3, it indicates that the cluster structure at this time is the most compact and the best separated.

(5) The Davies-Bouldin Score: It measures the average similarity between each cluster and its most similar cluster. Similarity is defined based on the intra-cluster divergence (size) and inter-cluster distance. When it is small, it means that the separation between the clusters is high and the inside of the cluster is the most compact.

Most criterion don't seem to find a optimal number of components, but the Calinski-Harabasz suggests that 2 might be a nice option.

2) K-Means: I also tried 4 scores on selection of k-value. For k-value, I didn't use BIC and AIC, instead, I used the simplest elbow chart to help us to find a k.

3 is also preferred by CH score, which also aligns with the number of our severity levels. I are not sure if they're relavant yet. And other scores didn't find an optimal k-value within this range.

C. Experiments on Dimensionality Reduction Algorithm

1) Randomized Projection: To determine the best number of components should the dataset have after dimensionality reduction, I used Reconstruction Error, a measurement that can tell us how many information can be retained after the transformation. Since I want to at least reduce some dimension without losing too much information, I drew a line where the dimensions left can avoid the error from the original data exceeding 5%. The visualization is the Fig.3, according to this measurement. So the smallest number of components I should preserve is 40.

I saved the transformed data after this process for later usage.

In order to get a rough idea of how this transformation does, I visualized the data points when the data is reduced

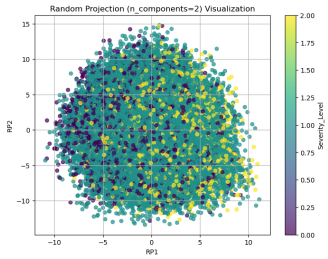

Fig. 4. Data Reduced to 2D with RP

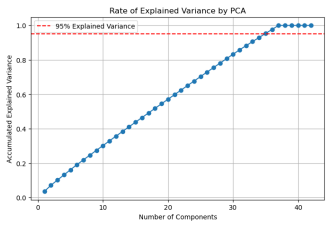

Fig. 5. Rate of Explained Variance vs. Number of Components for PCA

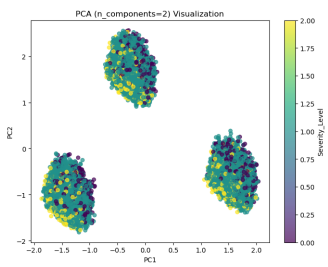

Fig. 6. Data Reduced to 2D with PCA

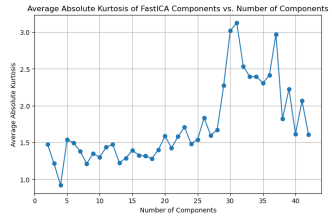

Fig. 7. Average Absolute Kurtosis of vs. Number of Components for ICA

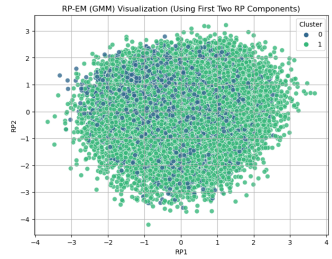

Fig. 8. Data Reduced to 2D with RP+EM

to 2 dimensions in Fig.4.

The label I define has three categories, which is shown in the corresponding color od 0, 1 and 2. Based on the graph, I can't really tell if the data is really transformed significantly, which makes sense because the projection process is randomized. But I noticed that the cluster 0 might be located more upleft and the cluster 2 is closer to downright, that might suggest that these two components are likely to be useful in the later classification task.

2) PCA: Like what I did with RP, to preserve at least 95% of the variance of data, I drew a line at that level on y-axis. The chart showing clearly where the least number I need is in Fig.5. And according to that chart, I should have 35 principle components using PCA.

I saved the transformed data after this process for later usage.

I also visualized the data when it's reduced to 2 dimensions, shown in Fig.5.

A good news that can be told from the graph is that by only looking at the first two components, the data can be clear divided into 3 clusters, in fact the real number can only be more considering there might exist some overlapping clusters that can't be seen in a 2D space. However, by the severity levels I defined is not the pattern I just found because the colors distribute similarly and equally in all three clusters. But since the distributions of colors in each cluster seems similar, I probably can divide them by using other dimensions.

3) ICA: Average Absolute Kurtosis is a good measurement of non-Gaussianity of components separated by ICA, so I tried to used its maxima as the optimal option of number of ICA components.

So based on Fig.7, the best number seems to be 31. I saved the transformed data after this process for later usage.

D. Experiments on Combinations

Now I can start using the dimensional reduced data with clustering algorithms to see how they turn out to be in combinations,

1) RP+EM: For the sake of space, I won't include every image of the same type in the following article

After dimensionality reduction using RP, Silhouette and SH scores both select 2 as the best number of components. It might be because RP didn't stretch or compress the data towards certain directions, so the data didn't change much besides has fewer features. It's normal that RP can not capture the real important structure of data.

In Fig.8, 2 clusters still look overlapping each other, suggesting the first 2 components might not be the most helpful directions in dividing them.

2) RP+K-Means: The CH score thinks 2 is the optimal number when I try K-Means on RP processed data, it is 1 cluster less than the outcome I got befor using RP. And this time the data projected into 2D space looks much clearer than the previous one, it can be clearly seen the data is classifiable with one cluster on the upleft and one cluster on the downright.

3) PCA+EM: This time Silhouette and Calinski-Harabasz scores select 7 and Davies-Bouldin selects 9. I incline to

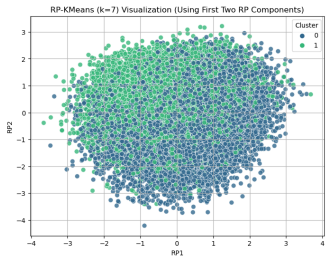

Fig. 9. Data Reduced to 2D with RP+K-Means

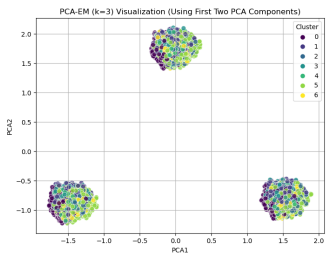

Fig. 10. Data Reduced to 2D with PCA+EM

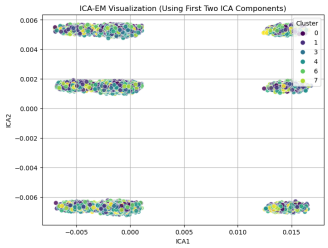

Fig. 12. Data Reduced to 2D with ICA+EM

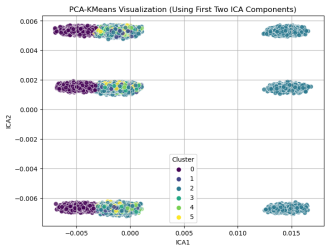

Fig. 13. Data Reduced to 2D with ICA+K-Means

follow the majority so I decide the optimal number to choose is 7.

Similar to the earlier result, the PCA does not create a pattern to clearly separate all the clusters in the first 2 components, but the relative positions of each cluster within the three groups that can be directly seen in the figure might convey some valuable information. (Fig. 9)

4) PCA+K-Means: CH score selects k=3 as the best number of components, so I visualized the data processed by PCA using K-Means.

This time the data is nicely clustered! (Fig.11) I reflected what I've gotten from previous steps, the reason why I didn't get good result using EM might not be EM's flaws. Selection of number of components plays an important role in this more than I thought. Because these PCA-treated clusters are predominantly circular profiles, K-Means did not hesitate to choose them, while EM had more concerns inside the more obvious clusters because they took more into account the Gaussian distribution and probability.

It can be concluded that Cancer dataset contains some components that is highly colinear and can be combined into 3 main clusters, PCA can do well in dimensionality reduction by linearly combining them into fewer components. And as shown in Fig.15, it can be nicely separated only by using first two principle components.

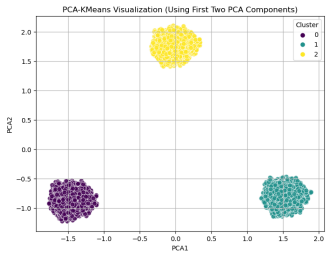

Fig. 11. Data Reduced to 2D with PCA+K-Means

5) ICA+EM: Silhouette and DB scores support 9 to be the optimal number, but in the Fig 12, things don't look well, I guess I encountered similar situation I had when using EM to PCA processed data.

6) ICA+K-Means: The 4 criterion couldn't find an optimal number this time, so based on the experience, I selected a number that seems to make sense according to the distribution I saw in Fig.12. And the outcome is shown in Fig.13.

This time the clusters don't have circular contour, that might be one of the reason why K-Means does not do well this time. There are two clusters clearly separated by one principle component while the other one doesn't look helpful, although these clusters seem far away from each other. I think this might one representation of curse of dimensionalities, that is to say although humans can tell these are clearly different clusters through visualization, It may be that in other dimensions, the distance between these clusters is not that large, so the distance may become relatively inconspicuous under the average of a large number of dimensions.

E. Experiments on NN

1) Unprocessed Data+NN: The accuracy in the training set is: 0.8424; The accuracy in the test set is: 0.8267, taking 4.8s.

The accuracy is not very impressive but the NN is not overfitting too much as well.

2) RP+NN: The accuracy in the training set is: 0.8452; The accuracy in the test set is: 0.8256, taking 5.6s.

The changes are minor, it performs similarly on RP processed data compared to the original one, and it's not running faster either.

3) PCA+NN: The accuracy in the training set is: 0.8524; The accuracy in the test set is: 0.0.8208, taking 7.4s.

PCA processed data is not running faster, and even shows a sign of overfitting. My guess is that NN models excel at dealing with nonlinear relationships. And PCA is a linear transformation. If there are complex nonlinear relationships in the raw data, PCA may linearize or lose these nonlinear information, making it more difficult for the neural network to capture these patterns, requiring more training iterations to converge.

4) ICA+NN: The accuracy in the training set is: 0.7396; The accuracy in the test set is: 0.7396, taking 3.5s.

ICA processed data significantly decrease training time by reducing number of features input to NN. But the accuracy seems to be sacrificed, which suggests ICA throws away too much information during processing. A very possible reason is that this dataset doesn't satisfy the basic assumptions of ICA, that is the real components lies within is not independent, but correlated. This idea actually makes sense because age, cancer stage and obesity might have a positive relation with each other, and so does features like countries and air pollution.

IV. EXPERIMENTS ON BANKRUPTCY DATASET

A. Data Exploration & Preprocessing

This dataset were collected from the Taiwan Economic Journal for the years 1999 to 2009. It contains 95 financial indicators commonly seen in financial statements like balance sheet, income statement and cash flow chart. It contains 6819 instances. Company bankruptcy was defined based on the business regulations of the Taiwan Stock Exchange.

Since it's acquired from Taiwan Stock Exchange, the label is already determined, whether this company is bankrupt. I don't need to construct a label by myself like I did in cancer dataset.

I standardized all features in this dataset. Each substracted their own means and divided by standard error.

B. Experiments on Clustering Algorithm

1) EM: The 4 crterions each got a different optimal number: BIC selects 4; Silhouette selects 2; Calinski-Harabasz selects 6 and Davies-Bouldin selects 7. Different metrics give different preferences, this might suggest that the original data is chaotic, it's hard to find good gaussian centers and standard errors. This might be a result of curse of dimensionalities.

2) K-Means: The elbow chart doesn't show a clear twist, and the other 3 criterions: Silhouette and DB score select 5; CH selects 3. I don't think any of these number is meaningful because I have too many features in the dataset, many financial ratios only have minor differences in their numerators or denominators, which means they're highly linear correlated.

C. Experiments on Dimensionality Reduction Algorithm

This dataset initially has 94 features.

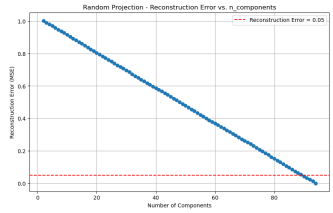

Fig. 14. Reconstruction Error vs. Number of Components for RP

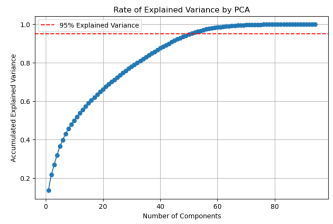

Fig. 15. Rate of Explained Variance vs. Number of Components for PCA

1) Randomized Projection: I did what I did on cancer dataset, as shown in Fig.14, I tried to reduce the number of dimensions without making an error that is larger than 5%, and the least number I need to include is 90.

So, the reduced number of features is 90.

2) PCA: While applying PCA, the metric is variance explained by the first n principle components as shown in Fig.15, the least number of PCs I got is 51, which is almost half of 94, it's a significant decrease in number of features train.

The reduced number of features is 51.

3) ICA: According to Fig.16, things are more direct with ICA, I can just use the number whose corresponding AAK reaches its maxima.

The reduced number of features is 49, this is more aggressive than PCA.

D. Experiments on Combinations

1) RP+EM: BIC selects 4 and Silhouette selects 2. To make reading figure easier I chose 2.

Fig.17 shows the data projected to 2D space and clustered with EM algorithm when number of clusters is 2. The distribution of data points looks radial. Maybe it's because some features have large variance and others have smaller one. Because all dimension in this graph is ramdomly generated, it doesn't tell us much how this algorithms perfoem.

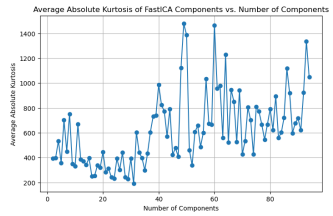

Fig. 16. Average Absolute Kurtosis of vs. Number of Components for ICA

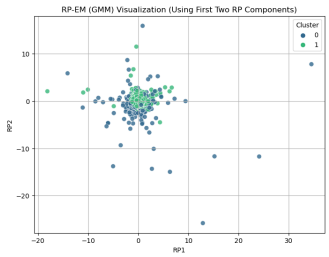

Fig. 17. Data Reduced to 2D with RP+EM

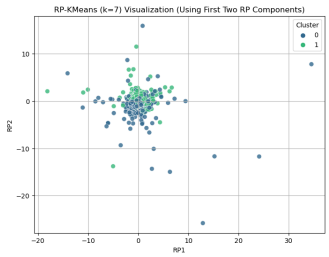

Fig. 18. Data Reduced to 2D with RP+K-Means

2) RP+K-Means: Silhouette selects 3 and Calinski-Harabasz selects 2.

To make Fig.18 comparable to Fig.17, I make its number of components 2 as well. The clustering results from these two algorithms have some subtle differences. EM seems to be more accepting of clusters overlapping each other, while K-Means is more inclined to separate clusters completely, which should be related to the probabilistic nature of EM's soft clustering, which can accept that a point has a certain probability of belonging to other clusters, which makes it more acceptable that points that are closer together are considered to belong to different clusters.

3) PCA+EM: Silhouette selects 2 and Davies-Bouldin selects 9, I prefer fewer number because too many colors add difficulties in finding structures.

4) PCA+K-Means: Silhouette selects 4, Calinski-Harabasz selects 3 and Davies-Bouldin selects 4.

Although I drew Fig 20 having 4 clusters, but the other two clusters has too few points, making the figure look like only having 2.

By comparing Fig.19 and Fig. 20, I can further vadidate what I thought about EM and K-Means. K-Means tries hard to make a clear cut wile EM tends to have clusters overlapping.

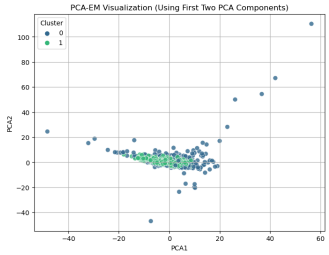

Fig. 19. Data Reduced to 2D with PCA+EM

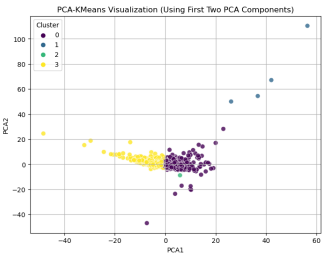

Fig. 20. Data Reduced to 2D with PCA+K-Means

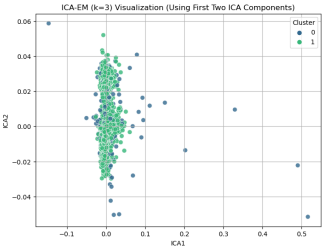

Fig. 21. Data Reduced to 2D with ICA+EM when k=2

5) ICA+EM: BIC/AIC selects 8 and Silhouette selects 2

6) ICA+K-Means: Silhouette/Davies-Bouldin selects 3.

And Fig.22 seems like only having 1 cluster, suggesting probably this dataset doesn't have individual components for ICA to separate. That's why the components are less helpful in clustering.

E. Experiments on NN

1) Unprocessed Data+NN: The accuracy in the training set is: 0.9868; The accuracy in the test set is: 0.9648, taking 1.8s.

2) RP+NN: The accuracy in the training set is: 0.9767; The accuracy in the test set is: 0.9626, taking 1.3s.

The NN trained RP processed data runs much faster thanks to reduction of dimensionality, and its accuracy doesn't change much.

3) PCA+NN: The accuracy in the training set is: 0.9677; The accuracy in the test set is: 0.9677, taking 0.5s.

The accuracy is improved a little and identical to the accuracy on training set, indicating the overfitting risk and curse of dimensionality is nicely solved by PCA. And benefiting from reduction of number of features, training time also shrinks to only 1/3 of the initial.

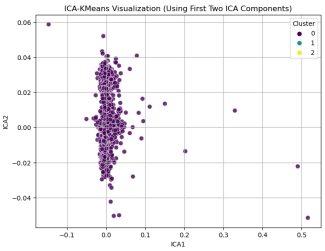

Fig. 22. Data Reduced to 2D with ICA+K-Means

4) ICA+NN: The accuracy in the training set is: 0.9692; The accuracy in the test set is: 0.9663, taking 0.7s.

It is possible that ICA discards some important information based on the assumption of component independence, which slows down the NN convergence speed, or that the features become sparse because more components are forcibly generated, thus slowing down the training speed.

But considering the accuracy is still pretty high, I incline to believe the latter. Maybe there're not so many independent components to separate, which makes the features sparse.

V. CONCLUSIONS

A. Dataset

1) Cancer Dataset: The assumption behind clustering algorithms, i.e., that K-Means or EM will be able to discern patient subpopulations, is validated to some degree. K-Means did show the ability for discrete cluster creation, especially on PCA-processed data, and even qualitatively matched the three severity levels I prescribed. The direct correspondence of these clusters to the true severity levels is not clear on the 2D map. EM algorithms, on the other hand, are struggling to decide on the optimal number of components and form overlapping clusters. That leads to the second hypothesis, i.e., EM will perform better than K-Means, which is not possible. In practice, K-Means performs better with clusters that will be less than likely to get elongated following PCA transformation, as EM is less "reluctant" due to its factor of taking into account Gaussian distribution and probability.

One of the assumptions about the algorithm, the assumption about the effectiveness of PCA stands true. PCA did succeed in reducing the dimensions while maintaining 95

2) Bankruptcy Dataset: Clustering attempts on raw high-dimensional data, whether EM or K-Means, have been futile to determine meaningful and interpretable clusters, reflecting the existence of a "dimensional curse." However, after dimensionality reduction, some clustering patterns began emerging. While K-Means works poorly in the original high-dimensional data in processing the linearity assumption of indicators, it beats EM in clustering partitioning following PCA dimension reduction, which proves the effectiveness of K-Means for linear separable forms.

My hypotheses on the effectiveness of the PCA are extremely valid. PCA could reduce 94 features to 51 principal components, and the principal components accounted for most of the variance, which is undoubtedly a radical feature reduction. On the random projection speed and performance hypotheses, RP does accelerate neural network training but isn't quite as effective at the dimensionality reduction effect (90 components required to maintain 95

B. Algorithms

Among all the clustering algorithms, K-Means performs best while attempting to find clusters that are separated and typically circular, especially after data preprocessing through linear dimension reduction such as PCA. It tends towards a "hard" classification of data points. EM, being a probabilistic method relying on a Gaussian mixture model, can support "soft" clustering. Its flexibility sometimes results in less distinct visual separations and the challenge of getting the proper number of components, especially when there isn't good dimensionality reduction or cluster shape isn't clearly Gaussian distributed. Amongst the methods of dimensionality reduction, PCA performed most robustly and effectively in this trial. It compresses dimensions effectively with a high percentage of variance in both sets. In high-dimensional bankruptcy data sets, PCA is the key to resisting the "dimensional curse", significantly improving the efficiency and generalization performance of neural networks. Its linear transformation nature is suitable in the event that there exists a clear linear correlation in data, such as financial indicators. Dataset reduction in low-dimensional datasets can be accomplished using cross-validation. Random projection (RP), although faster, is not as efficient as PCA as a method of reducing dimensions, and offers only a small advantage to high-dimensional datasets for training time. The poor performance of ICA with the cancer database shows that the components of the database may not be independent and thus cause information loss and reduced neural network accuracy. As much as ICA does show effective dimensionality reduction and improve neural network efficiency on bankruptcy databases, its imposition of independence can result in sparse features and hence its overall performance slightly below that of PCA. The combination of neural network efficiency and dimensionality reduction reveals the decisive influence of the features of the dataset on its effect. For high-dimensional and highly intercorrelated datasets for bankruptcy, dimensionality reduction, especially PCA, is extremely beneficial, dramatically cutting down training time, and improving accuracy and generalization capabilities. This strongly proves that feature reduction in this context is an optimal solution to solve the dimensional curse and support the learner. However, for

For similarly low-dimensional cancer data sets, the benefit of dimensionality reduction on neural network performance is uncertain and can even lead to an accuracy decrease (as shown by PCA and ICA). This proves that for small-sized data sets, linear dimensionality reduction is not always going to improve neural network performance, especially when nonlinear relationships are of utmost significance.